Detecting and Recovering from Deception in Model-Free Nash-Equilibrium Seeking: A Dual-Demodulation Analysis

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Abstract

We give a self-contained mathematical justification of a model-free deception detector for Nash-equilibrium (NE) seeking dynamics. The detector is built *entirely* from the demodulation signal C_v that each player already computes. We prove three facts. First, a *blindness* lemma: the raw signal C_v vanishes at the deceptive equilibrium exactly as it does at an honest NE, so C_v alone cannot certify honesty. Second, a *detection* theorem: a second demodulation C_v' taken at a probe frequency unknown to the deceiver yields the discrepancy $D_v = |C_v' - C_v|$, which is identically zero under honest play and bounded below by an *explicit* positive constant whenever deception is active, giving a clean separation threshold (no false alarms, no missed detections). Third, a *recovery* theorem: re-randomizing the victim's probe frequency collapses the deceptive equilibrium and restores exponential convergence to the true NE. The development uses only smoothness, standard averaging, and a single non-degeneracy condition that characterizes *effective* deception.

1 Setup and standing assumptions

We consider two players, a *victim* v and a *deceiver* d, with scalar actions $x_v, x_d \in \mathbb{R}$ and twice-continuously differentiable cost functions $J_v(x_v, x_d)$ and $J_d(x_v, x_d)$. (The vector / N-player extension is described in Remark 4.) Each player runs the model-free NE-seeking (extremum-seeking) dynamics

$$x_v(t) = u_v(t) + a\sin(\omega_v t), \qquad \dot{u_v} = -\tfrac{2k}{a}\, J_v\big(x(t)\big)\sin(\omega_v t), \tag{1}$$

$$x_d(t) = u_d(t) + a\sin(\omega_d t) + a\,\delta_d(t)\sin(\omega_v t), \qquad \dot{u_d} = -\tfrac{2k}{a}\, J_d\big(x(t)\big)\sin(\omega_d t), \tag{2}$$

where $a > 0$ is the probing amplitude, $k > 0$ a gain, and $x = (x_v, x_d)$. The deceiver additionally injects the victim's probe $\sin(\omega_v t)$ into its own action with a time-varying amplitude δ_d updated by integral feedback toward a desired payoff J_d^{ref}:

$$\dot{\delta_d} = \varepsilon\big(J_d(x) - J_d^{\mathrm{ref}}\big), \qquad \varepsilon > 0. \tag{3}$$

When $\delta_d \equiv 0$ the system reduces to ordinary (honest) NE seeking. Throughout, $\partial_v J_v := \partial J_v/\partial x_v$, $\partial_d J_v := \partial J_v/\partial x_d$, etc., are evaluated at the slow state $u = (u_v, u_d)$, and $\overline{(\cdot)}$ denotes the infinite-horizon time average $\overline{f} := \lim_{T\to\infty} \frac{1}{T}\int_0^T f(t)\,dt$ taken with u, δ_d held frozen.

We keep the assumptions minimal.

Assumption 1 (Regularity)**.** $J_v, J_d \in C^3$ on an open operating set $\mathcal{U} \subset \mathbb{R}^2$ containing the trajectories of interest.

Assumption 2 (Probe selection)**.** The probing frequencies $\omega_v, \omega_v', \omega_d > 0$ used below are distinct and *non-resonant*: no integer combination $m_1\omega_v + m_2\omega_v' + m_3\omega_d$ with $|m_1| + |m_2| + |m_3| \le 3$ vanishes except the trivial one. Moreover $\min\{\omega_v, \omega_v', \omega_d\}$ is large enough that averaging applies (time-scale separation between the probes and the slow dynamics (1)–(3)).

Assumption 2 is exactly the orthogonality condition that makes extremum seeking work; it guarantees the elementary averages

$$\overline{\sin(\omega_i t)\sin(\omega_j t)} = \tfrac{1}{2}\mathbb{1}[\omega_i = \omega_j], \qquad \overline{\sin(\omega_i t)\sin(\omega_j t)\sin(\omega_\ell t)} = 0, \tag{4}$$

for any probe frequencies appearing in (1)–(2).

By standard averaging theory [3, Ch. 10], the trajectories of the true (time-varying) system (1)–(3) remain $O(a)$-close, uniformly on compact time intervals and—under the stability conditions invoked in Theorem 2—uniformly in t, to those of the averaged system obtained below. We therefore carry out the analysis on the averaged system and record the residual probing bias as $O(a^2)$; all conclusions transfer to the real system up to $O(a)$.

2 Averaged dynamics and the deceptive equilibrium

The single object underlying everything is the *demodulation* of the victim's payoff at a frequency ω,

$$\mathcal{D}_\omega[J_v] \;:=\; \overline{\tfrac{2}{a}\, J_v(x)\, \sin(\omega t)}. \tag{5}$$

This is precisely the quantity the victim already forms inside its update (1): indeed $\dot{u}_v = -k\, \mathcal{D}_{\omega_v}[J_v]$ on average.

Lemma 1 (Demodulation recovers a perceived gradient)**.** *Under Assumptions 1–2, with the deceiver injecting $a\delta_d \sin(\omega_v t)$ as in (2):*

(a) *Demodulating at the victim's operating frequency ω_v,*

$$C_v := \mathcal{D}_{\omega_v}[J_v] = \partial_v J_v(u) \;+\; \delta_d\, \partial_d J_v(u) \;+\; O(a^2). \tag{6}$$

(b) *If the victim additionally probes with amplitude a at a frequency $\omega_v' \notin \{\omega_v, \omega_d\}$* not used by the deceiver*, then demodulating at ω_v',*

$$C_v' := \mathcal{D}_{\omega_v'}[J_v] = \partial_v J_v(u) \;+\; O(a^2). \tag{7}$$

Proof. Write the action perturbations $x_v - u_v = a\sin(\omega_v t)$ ($+\, a\sin(\omega_v' t)$ in case (b)) and $x_d - u_d = a\sin(\omega_d t) + a\delta_d \sin(\omega_v t)$. A third-order Taylor expansion of J_v about u gives

$$J_v(x) = J_v(u) + \partial_v J_v\,(x_v - u_v) + \partial_d J_v\,(x_d - u_d) + R_2, \qquad R_2 = O(a^2),$$

where R_2 collects the second- and third-order terms.

Case (a). Multiply by $\frac{2}{a}\sin(\omega_v t)$ and average. The constant term gives $\frac{2}{a} J_v(u)\overline{\sin(\omega_v t)} = 0$. By (4),

$$\tfrac{2}{a}\,\partial_v J_v\,\overline{(x_v - u_v)\sin(\omega_v t)} = \tfrac{2}{a}\,\partial_v J_v \cdot a\,\tfrac{1}{2} = \partial_v J_v,$$

$$\tfrac{2}{a}\,\partial_d J_v\,\overline{(x_d-u_d)\sin(\omega_v t)} = \tfrac{2}{a}\,\partial_d J_v\,\Big(a\,\underbrace{\overline{\sin(\omega_d t)\sin(\omega_v t)}}_{=0} + a\delta_d\,\underbrace{\overline{\sin(\omega_v t)\sin(\omega_v t)}}_{=1/2}\Big) = \delta_d\,\partial_d J_v.$$

Every second-order term of R_2 is a product of *two* probe sinusoids times the extra $\sin(\omega_v t)$, i.e. a product of three sinusoids, and averages to zero by (4); the third-order remainder is $O(a^3)$ and contributes $\frac{2}{a}\cdot O(a^3) = O(a^2)$. This proves (6).

Case (b). Multiply instead by $\frac{2}{a}\sin(\omega_v' t)$ and average. Now the $\partial_v J_v$ term yields $\partial_v J_v$ exactly as above (the victim's own ω_v'-probe). The deceiver injects *nothing* at ω_v', so

$$\overline{(x_d-u_d)\sin(\omega_v' t)} = \overline{\big(a\sin(\omega_d t) + a\delta_d\sin(\omega_v t)\big)\sin(\omega_v' t)} = 0$$

because $\omega_v' \neq \omega_d$ and $\omega_v' \neq \omega_v$. Hence the $\partial_d J_v$ contribution drops and (7) follows, the remainder again being $O(a^2)$. □

Remark 1. Equation (6) is the mechanism behind deception: the victim's update $\dot{u}_v = -k\,C_v$ drives u_v to a point where the *perceived* gradient $\partial_v J_v + \delta_d\,\partial_d J_v$, not the true gradient $\partial_v J_v$, vanishes.

The averaged closed loop is therefore

$$\dot{u}_v = -k\big(\partial_v J_v(u) + \delta_d\,\partial_d J_v(u)\big) + O(a^2), \quad \dot{u}_d = -k\,\partial_d J_d(u) + O(a^2), \quad \dot{\delta}_d = \varepsilon\big(J_d(u) - J_d^{\mathrm{ref}}\big) + O(a^2). \tag{8}$$

Definition 1 (Honest NE and deceptive NE)**.** A point $u^\star$ is a *(true) Nash equilibrium* if $G_{\mathrm{hon}}(u^\star) = 0$, where the honest pseudogradient is $G_{\mathrm{hon}}(u) := \big(\partial_v J_v(u),\, \partial_d J_d(u)\big)$. A triple (u^*, δ_d^*) with $\delta_d^* \neq 0$ is a *deceptive Nash equilibrium* (DNE) if it is an equilibrium of (8), i.e.

$$\partial_v J_v(u^*) + \delta_d^*\,\partial_d J_v(u^*) = 0, \qquad \partial_d J_d(u^*) = 0, \qquad J_d(u^*) = J_d^{\mathrm{ref}}. \tag{9}$$

The first identity in (9) is the crux of the whole analysis: it says

$$\boxed{\partial_v J_v(u^*) = -\,\delta_d^*\,\partial_d J_v(u^*)} \tag{$\star$}$$

so the victim's *true* gradient at the DNE is generically *nonzero*, equal in magnitude to $|\delta_d^*|\,|\partial_d J_v(u^*)|$.

3 Why the raw signal C_v cannot detect deception

Lemma 2 (Single-frequency blindness)**.** *Along* (8)*, the demodulation satisfies* $C_v = -\frac{1}{k}\dot{u}_v + O(a^2)$*. Consequently* $C_v = O(a^2)$ *at* every *equilibrium of the victim's slow dynamics—in particular at an honest NE* ($\delta_d = 0$) and *at a DNE* ($\delta_d = \delta_d^* \neq 0$)*. Hence no test applied to* C_v *alone can distinguish honest convergence from deceptive convergence.*

Proof. The identity $C_v = -\frac{1}{k}\dot{u}_v + O(a^2)$ is (8). At an honest NE, $\delta_d = 0$ and $\partial_v J_v(u^\star) = 0$, so $C_v = O(a^2)$. At a DNE, by the first relation in (9) the perceived gradient is zero, so $\dot{u}_v = O(a^2)$ and again $C_v = O(a^2)$. The two limits coincide, so C_v carries no information separating the two regimes. □

Lemma 2 is exactly why an action-based residual $r_v = x_v(t{+}1) - \hat{x_v}(t{+}1)$ (the difference between the realized action and the action predicted by the honest update rule) also fails: the victim *does* follow the honest rule; only its *input* J_v is corrupted, so r_v is $O(a^2)$ by construction. The defect is not the residual idea but the use of a *single* compromised frequency.

4 A detector built from a second demodulation

The fix uses nothing beyond a second copy of the same demodulation (5), taken at a frequency ω_v' the deceiver does not know and hence cannot inject into. Define

$$\boxed{D_v := \left| C_v' - C_v \right|} \tag{10}$$

with C_v, C_v' as in Lemma 1.

Theorem 1 (Detection identity)**.** *Under Assumptions 1–2, along the averaged dynamics (8),*

$$D_v \;=\; |\delta_d| \left|\partial_d J_v(u)\right| \;+\; O(a^2), \tag{11}$$

at every time t (not only at steady state).

Proof. Subtract (7) from (6):

$$C_v' - C_v = \big(\partial_v J_v(u) + O(a^2)\big) - \big(\partial_v J_v(u) + \delta_d\, \partial_d J_v(u) + O(a^2)\big) = -\,\delta_d\, \partial_d J_v(u) + O(a^2).$$

Taking absolute values gives (11). The true gradient $\partial_v J_v(u)$ appears in both demodulations and cancels; what survives is precisely the injection term that (6) could not isolate. □

To turn (11) into a decision rule we use one non-degeneracy condition, which simply states that deception is *effective*—if it failed, there would be nothing to detect.

Assumption 3 (Effective deception)**.** There is a constant $c_0 > 0$ with $|\partial_d J_v(u)| \geq c_0$ for all $u \in \mathcal{U}$. We say deception is *active* at time t if $|\delta_d(t)| \geq \delta_{\min}$ for a fixed $\delta_{\min} > 0$.

Corollary 1 (Soundness, completeness, separation)**.** *Let Assumptions 1–3 hold, and choose a small enough that the $O(a^2)$ term in (11) is bounded by $\frac{1}{4}\delta_{\min} c_0$ in magnitude on $\mathcal{U}$. Fix the threshold*

$$\rho := \tfrac{1}{2}\, \delta_{\min}\, c_0.$$

Then:

(i) ***Soundness (no false alarm).*** *Under honest play ($\delta_d \equiv 0$), $D_v \leq \frac{1}{4}\delta_{\min} c_0 < \rho$ for all t.*

(ii) ***Completeness (no missed detection).*** *Whenever deception is active,*

$$D_v \;\geq\; \delta_{\min} c_0 - \tfrac{1}{4}\delta_{\min} c_0 \;=\; \tfrac{3}{4}\, \delta_{\min} c_0 \;>\; \rho.$$

(iii) ***Separation.*** *Consequently the rule*

$$D_v > \rho \implies \textit{deception active}, \qquad D_v \leq \rho \implies \textit{honest},$$

is correct: it produces neither false alarms nor missed detections.

Proof. For (i), $\delta_d \equiv 0$ in (11) gives $D_v = O(a^2)$, bounded by $\frac{1}{4}\delta_{\min} c_0$ by the choice of a. For (ii), Assumption 3 gives $|\delta_d||\partial_d J_v| \geq \delta_{\min} c_0$, and subtracting the $O(a^2)$ bound yields the stated lower bound, which exceeds ρ. Part (iii) is immediate from (i)–(ii), since the honest range $[0, \rho)$ and the deceptive range (ρ, ∞) are disjoint. □

Remark 2. The detector D_v uses only the victim's own payoff samples J_v and its own self-generated probes at ω_v, ω_v'; it requires no model of J_v, no knowledge of the deceiver's identity, and no knowledge of δ_d or J_d. It is therefore admissible in the same model-free setting as the NE-seeking dynamics themselves.

5 Recovery: collapsing the deceptive equilibrium

Detection and recovery use the *same* primitive—a fresh probe frequency unknown to the deceiver. Suppose at a trigger time t_0 (when $D_v > \rho$) the victim re-randomizes the frequency that drives its *update* (1) from ω_v to a new secret value $\omega_v'' \notin \{\omega_v, \omega_d\}$, also satisfying Assumption 2. We use one stability hypothesis on the honest game.

Assumption 4 (Stable true NE)**.** The honest pseudogradient flow $\dot{u} = -k\, G_{\text{hon}}(u)$ has an equilibrium $u^\star$ (Definition 1) at which $-k\, DG_{\text{hon}}(u^\star)$ is Hurwitz. (This holds, e.g., for strongly monotone games.)

Theorem 2 (Recovery to the true NE)**.** *Let Assumptions 1, 2, 4 hold. After the switch $\omega_v \mapsto \omega_v''$ at t_0, there exist $\bar{a}, \bar{\varepsilon} > 0$ and $\bar{\omega} < \infty$ such that whenever $a \in (0, \bar{a})$, $\varepsilon \in (0, \bar{\varepsilon})$, and the probe frequencies exceed $\bar{\omega}$, the slow state $u(t) = (u_v(t), u_d(t))$ of (1)–(2) converges exponentially to an $O(a)$-neighborhood of the true NE $u^\star$,* for every *trajectory of $\delta_d(t)$. In particular the victim's action satisfies $x_v(t) \to u_v^\star + a\sin(\omega_v'' t) + O(a)$.*

Proof. After the switch the victim demodulates and updates at ω_v'', while the deceiver—ignorant of ω_v''—still injects only at the old frequency ω_v. Repeating the computation of Lemma 1(a) with ω_v replaced by ω_v'' in the update and the injection left at ω_v, the cross term now carries the factor $\sin(\omega_v t)\sin(\omega_v'' t) = 0$ by (4). Hence the perceived gradient loses its bias and the averaged victim dynamics revert to the honest law:

$$\dot{u_v} = -k\,\partial_v J_v(u) + O(a^2), \qquad \dot{u_d} = -k\,\partial_d J_d(u) + O(a^2),$$

i.e. $\dot{u} = -k\, G_{\text{hon}}(u) + O(a^2)$. Crucially, δ_d *no longer appears* in the u-dynamics: the deceiver's integral state (3) is decoupled from u, so whatever $\delta_d(t)$ does is irrelevant to the victim's convergence.

By Assumption 4, $u^\star$ is a locally exponentially stable equilibrium of $\dot{u} = -k\, G_{\text{hon}}(u)$; let V be a quadratic Lyapunov function with $\dot{V} \le -c\|u - u^\star\|^2$ near $u^\star$. The $O(a^2)$ probing bias is a vanishing perturbation, so for a small enough u converges exponentially to an $O(a^2)$-ball around $u^\star$ on the averaged system. By the averaging theorem [3, Thm. 10.4], this exponential stability of the averaged system implies the trajectories of the true time-varying system (1)–(2) converge exponentially to an $O(a)$-neighborhood of $u^\star$, provided the probe frequencies are sufficiently large; the small-ε requirement places the decoupled slow integrator (3) in standard singular-perturbation form [3, Ch. 11] and does not affect the u-limit. This is the claimed result. □

Remark 3. Theorem 2 explains the asymmetry the deceiver relies on: its influence channel is the single frequency ω_v, and that channel is destroyed the instant the victim moves to a frequency the deceiver cannot observe. The detection statistic D_v and the recovery action are two uses of the same secret-probe idea—one reads the bias, the other removes it.

Remark 4 (Vector and N-player extension)**.** For $x_v \in \mathbb{R}^{n_v}$ and several deceivers, replace $\partial_v J_v, \partial_d J_v$ by the corresponding Jacobian blocks $\nabla_v J_v$, $\nabla_d J_v$, the scalar δ_d by the injection-amplitude vector δ, and absolute values by Euclidean norms. Lemma 1 holds componentwise, (11) becomes $D_v = \|(\nabla_d J_v)\,\delta\| + O(a^2)$, Assumption 3 becomes the coercivity bound $\|(\nabla_d J_v)\delta\| \ge c_0\|\delta\|$ on $\mathcal{U}$, and Assumption 4 becomes Hurwitzness of $-k\, DG_{\text{hon}}(u^\star)$ (implied by strong monotonicity of the honest game). The statements and proofs of Theorem 1, Corollary 1, and Theorem 2 carry over verbatim.

Summary of the logical chain

1. Lemma 1: each demodulation returns a perceived gradient; the compromised one carries the injection term, the secret one does not.
2. Lemma 2: hence the raw C_v vanishes at the DNE just as at an honest NE—it is blind.
3. Theorem 1 + Corollary 1: the discrepancy $D_v = |C'_v - C_v|$ equals $|\delta_d||\partial_d J_v| + O(a^2)$, giving a sound, complete, threshold-separated detector.
4. Theorem 2: switching the probe frequency decouples δ_d from the victim and restores exponential convergence to the true NE.

References

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